4 Compounds **P** and **Q** are structural isomers.



Fig. 4.1

(a) (i) Define structural isomerism.

.....

..... [2]

(b) Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

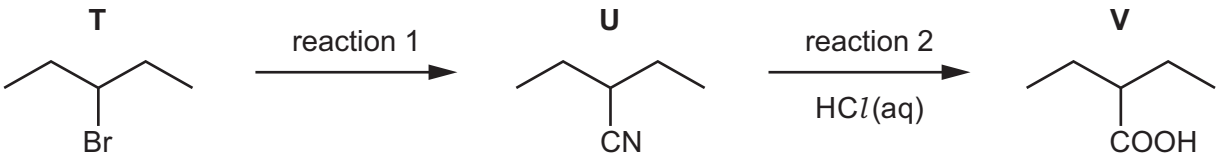


Fig. 4.3

(i) Identify the reagent and conditions for reaction 1.

..... [1]

(iii) **V** reacts with propan-2-ol in the presence of a catalytic amount of H_2SO_4 to form organic compound **W**.

Complete Table 4.1 to give details of this reaction.

Table 4.1

	reaction of **V** with propan-2-ol
type of reaction	
functional group formed	
molecular formula of organic product **W**	

[3]

Table 3.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

[Total: 13]

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6.

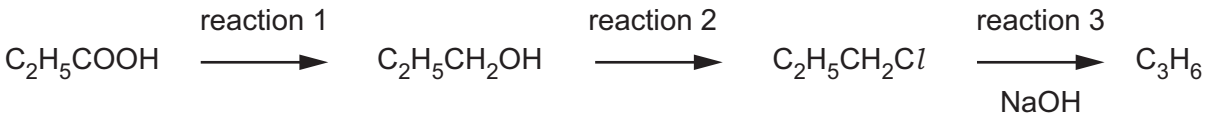


Fig. 4.1

(a) (i) Identify the type of reaction that occurs in reaction 1.

..... [1]

4 Fig. 4.1 shows how propane, C_3H_8, can be converted to propanoic acid, CH_3CH_2COOH.

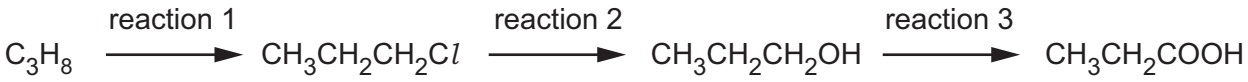


Fig. 4.1

(a) Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.

$$C_3H_8 + Cl_2 \rightarrow CH_3CH_2CH_2Cl + HCl$$

The reaction takes place via initiation, propagation and termination steps.

(i) Name the mechanism shown by reaction 1.

..... [1]

(d) CH_3CH_2COOH reacts with an unsaturated alcohol **R** to form unsaturated ester **S**.

(i) State the **type** of reaction that forms **S**.

..... [1]

Table 3.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

[Total: 13]

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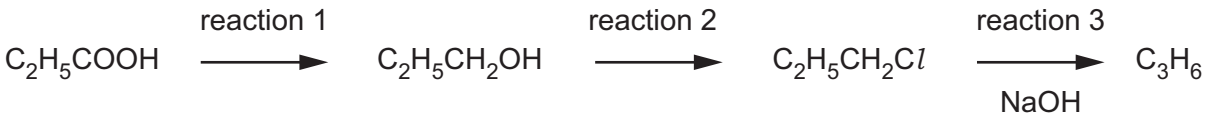


Fig. 4.1

(a) (i) Identify the type of reaction that occurs in reaction 1.

..... [1]

4 The skeletal formula of **E** is shown in Fig. 4.1.

E

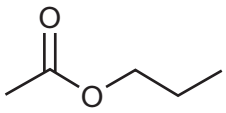


Fig. 4.1

(a) State the molecular formula of **E**.

..... [1]

(b) Name **E**.

..... [1]

(c) **E** is made when a carboxylic acid and an alcohol react together.

(i) Name the type of reaction that takes place during this esterification reaction.

..... [1]

(ii) Write an equation for the formation of **E** from a carboxylic acid and an alcohol. Use structural formulae to represent the organic species.

..... [1]

(iii) State the systematic name of the alcohol used to make **E**.

..... [1]

(d) **F** is an isomer of **E**.
F reacts with HC*l*(aq) to produce **G** and **H**.
G is a secondary alcohol.
H is also produced when HC*l*(aq) is added to CH_3CN.

(i) Construct an equation for the acid hydrolysis of CH_3CN with HC*l*(aq).

..... [1]

(ii) Draw the displayed formula of **G**.

[2]

5 **(a)** **W** is a colourless liquid.

W

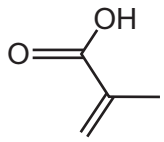


Fig. 5.1

(i) Deduce the empirical formula of **W**.

..... [1]

(ii) **W** reacts with $LiAlH_4$ to produce **Y**.

Draw the structure of **Y**.

[1]

(iii) Identify the role of $LiAlH_4$ when it reacts with **W**.

(iv) Suggest the reagent and conditions required when **W** is converted into **Z**.

..... [1]

4 **(a)** Propanoic acid, CH_3CH_2COOH, reacts with reducing agent **Q** to produce propan-1-ol, $CH_3CH_2CH_2OH$.

(i) Suggest the formula of reducing agent **Q**.

..... [1]

(ii) Complete the equation to show the reduction of propanoic acid to propan-1-ol.
Use [H] to represent one atom of hydrogen from **Q**.

.....CH_3CH_2COOH + ...[H] $\rightarrow$ ..... [1]

[Total: 9]

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C−O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C−H	alkane	2850–2950
N−H	amine, amide	3300–3500
O−H	carboxyl hydroxy	2500–3000 3200–3650

5 Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $H_2SO_4(aq)$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

$$\mathbf{A} + 2H_2O \xrightarrow[\text{heat}]{H_2SO_4(aq)} \underset{\mathbf{B}}{HOOCCH_2COOH} + \underset{\mathbf{C}}{2CH_3CH_2OH}$$

Fig. 5.1

(a) Draw the structure of **A**.

[2]

(b) Name the type of reaction that occurs when **A** is heated with $H_2SO_4(aq)$.

..... [1]

(c) When aqueous Na_2CO_3 is added to separate samples of **B** and **C**, effervescence is observed with **B** only.

(i) Complete the equation to describe the reaction between **B** and an excess of aqueous Na_2CO_3.

.....$HOOCCH_2COOH$ + ...Na_2CO_3 → .....
[1]

(e) **B** reacts with an excess of reducing agent **R** to produce **D**.

An excess of PCl_5 is added to **D**.

- A vigorous reaction occurs.
- Misty fumes are seen.
- Organic compound **E** is produced.

(i) Identify the formula of the reducing agent **R**.

..... [1]

(ii) Complete the equation to describe the reaction of **B** with an excess of **R**.
Use [H] to represent one atom of hydrogen from **R**.

.....$HOOCCH_2COOH$ + ...[H] → ..... [2]

4 Propanone, CH_3COCH_3, is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

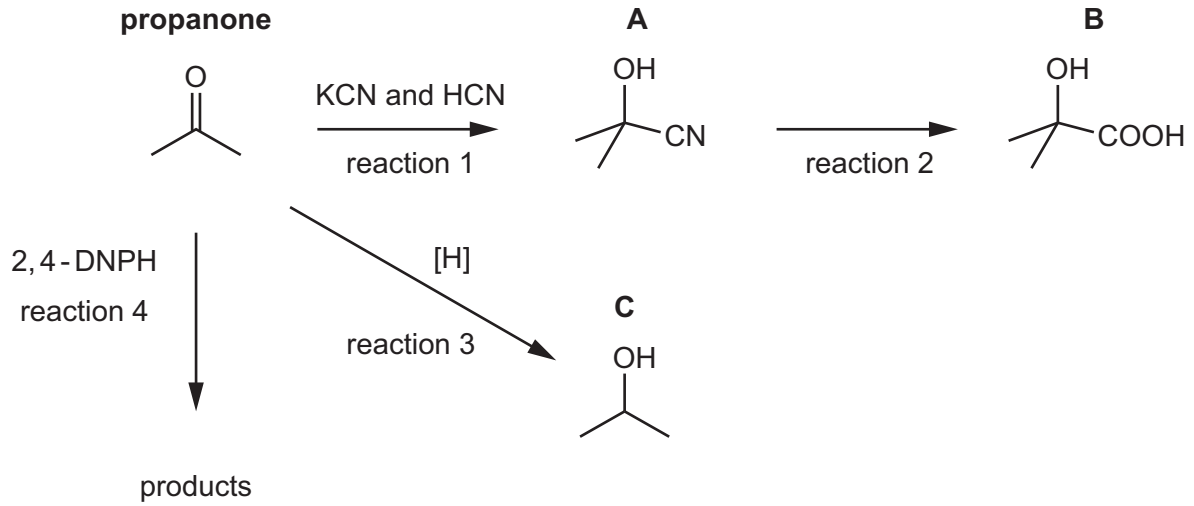


Fig. 4.1

(a) Reaction 1 is a nucleophilic addition reaction.

(i) Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(b) Suggest the reagents and conditions for reaction 2.

..... [1]

4 Bromoalkanes are used widely in industry, although there is increasing concern about their environmental impact.

Fig. 4.1 shows a reaction scheme involving 1,2-dibromoethane.

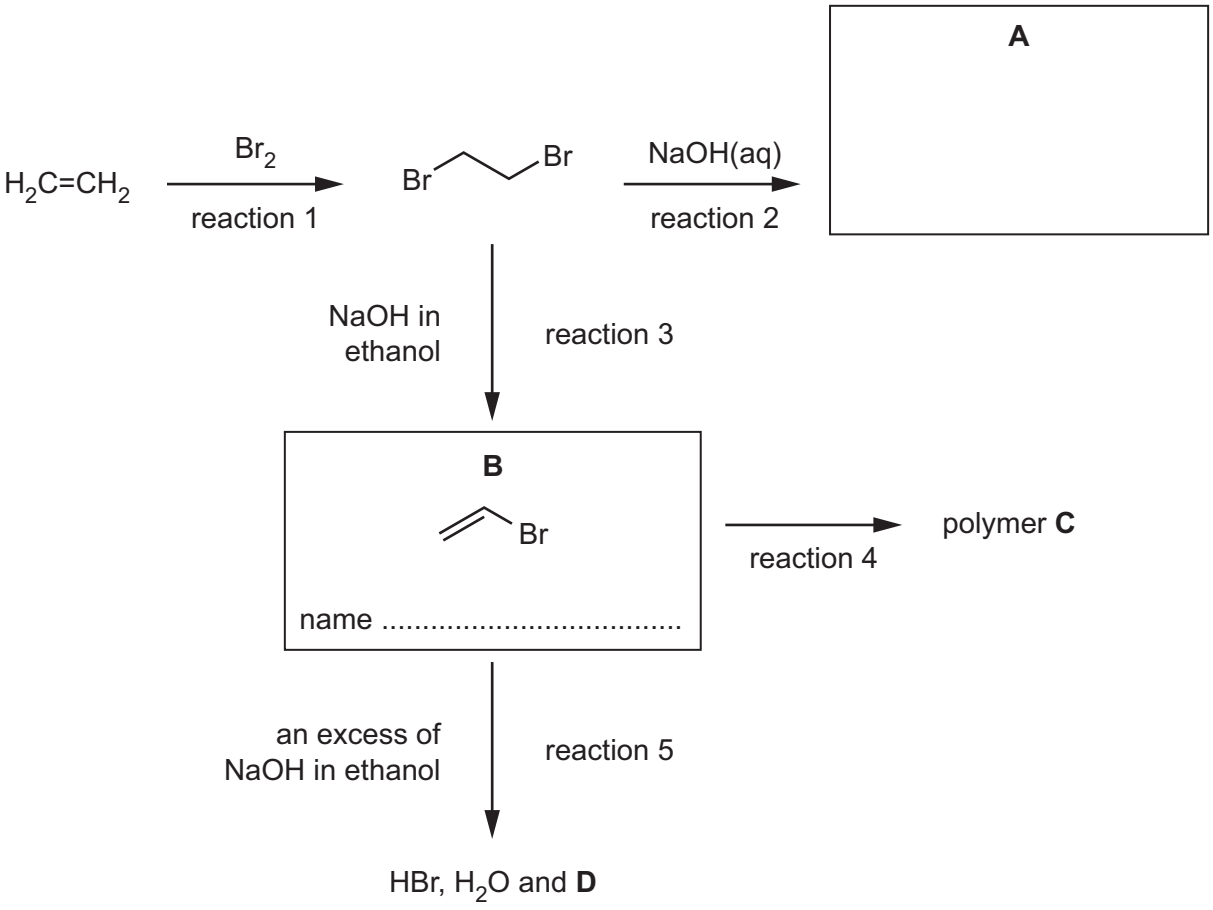


Fig. 4.1

(a) Complete Fig. 4.2 to show the mechanism for the formation of 1,2-dibromoethane in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



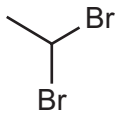
Fig. 4.2

[3]

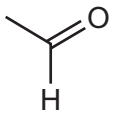
(d) Compound **E** is the only isomer of 1,2-dibromoethane.

Alkaline hydrolysis of **E** gives compound **F**.

E



F



(i) Identify the type of isomerism shown by **E** and 1,2-dibromoethane.

..... [1]

Question 4 continues on page 14.

(e) Compound **F** reacts with reagent **G** to form compound **H**.



The infrared spectrum of **H** is shown in Fig. 4.3.

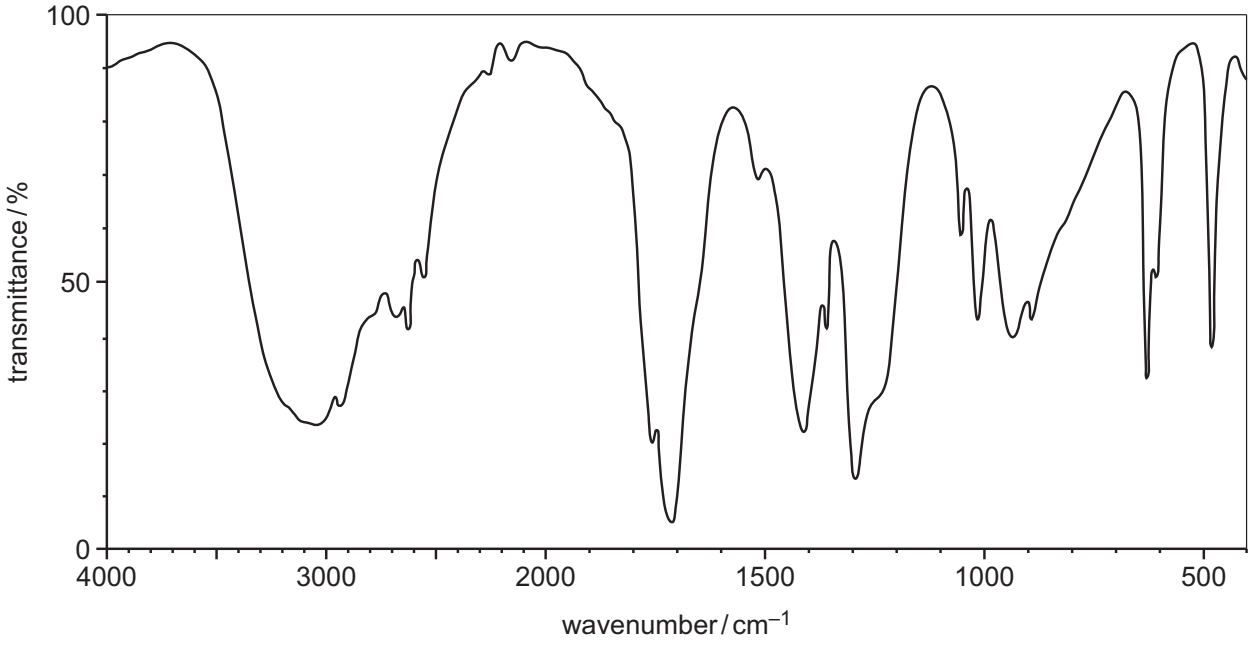


Fig. 4.3

Table 4.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

H also shows a molecular ion peak at *m*/*e* = 60 in its mass spectrum.

(i) Use the information in **(e)**, Fig. 4.3 and Table 4.2 to deduce the structure of **H**. Explain your answer fully.

H

.....

.....

.....

[3]

(ii) Suggest the role of reagent **G**.

.....

[1]

4 Propanone, CH_3COCH_3, is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

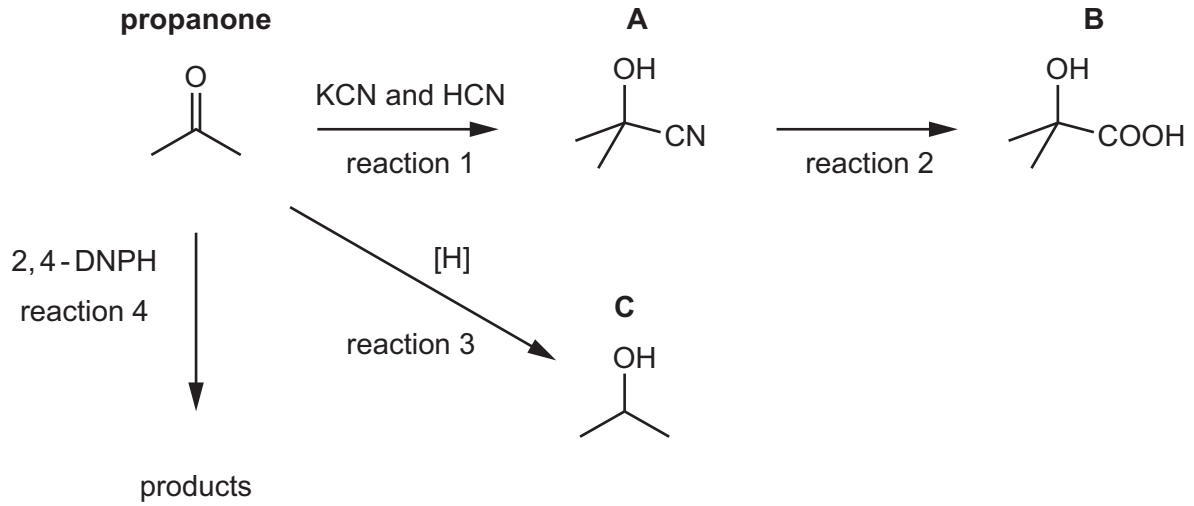


Fig. 4.1

(a) Reaction 1 is a nucleophilic addition reaction.

(i) Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(b) Suggest the reagents and conditions for reaction 2.

..... [1]

5 **(a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

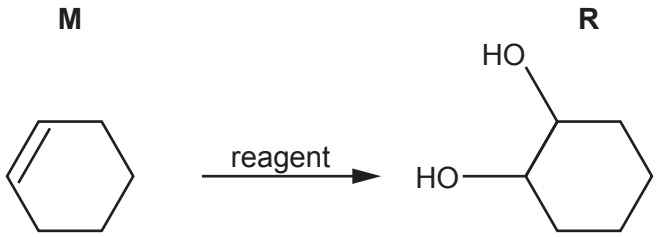


Fig. 5.1

Identify the reagent and conditions for this reaction.

..... [1]

(d) **Y** is made from **Q** in a three-step reaction.

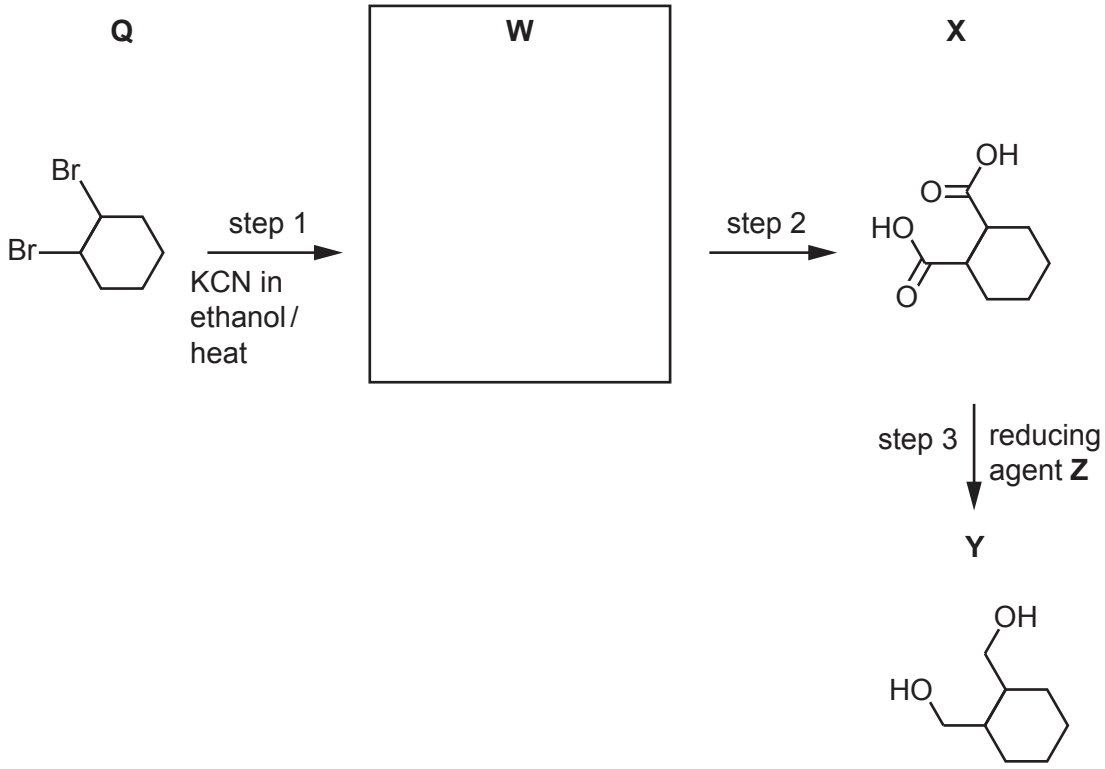


Fig. 5.4

(i) Draw the structure of **W** in the box in Fig. 5.4. [1]

(ii) In step 2, **W** is heated with HCl(aq) to produce **X** and an inorganic product.

Identify the formula of the inorganic product.

..... [1]

(iii) In step 3, **X** reacts with reducing agent **Z** to produce **Y**.

Complete the equation for the reaction of **X** with **Z**.

Use a molecular formula to represent the organic product.

Use [H] to represent one atom of hydrogen from **Z**.

…… $C_8H_{12}O_4$ + ……[H] ⟶ ..... [1]

(iv) Identify **Z**.

..... [1]

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

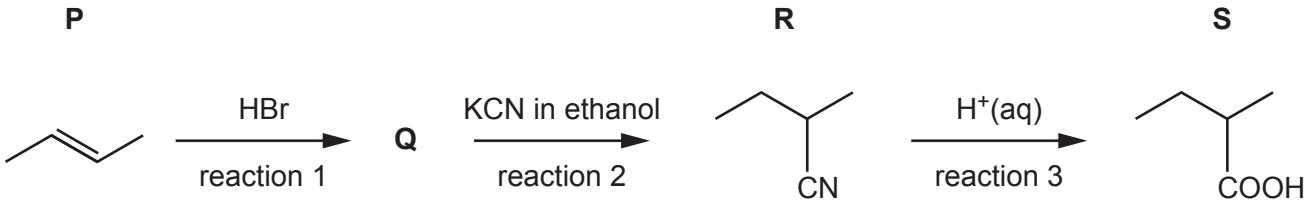


Fig. 4.1

(a) **P**, **Q**, **R** and **S** show stereoisomerism.

Complete Table 4.1 by identifying with a tick (✓) the type of stereoisomerism that each molecule shows.

The type of stereoisomerism shown by **Q** is given.

Table 4.1

	P	**Q**	**R**	**S**
geometrical isomerism				
optical isomerism		✓		

[2]

(c) Compounds **S** and **T** react to form organic compound **U**, which has a single functional group.

S

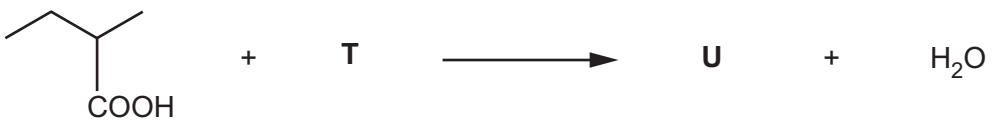


Table 4.2 shows some data from the mass spectrum of **U**.

Table 4.2

peak	relative abundance
M^+	7.2
$[M+1]^+$	0.55

(i) Use the data from Table 4.2 to show that **U** contains 7 carbon atoms.

Show your working.

[2]

(ii) Fig. 4.2 shows the infrared spectrum of **U**.

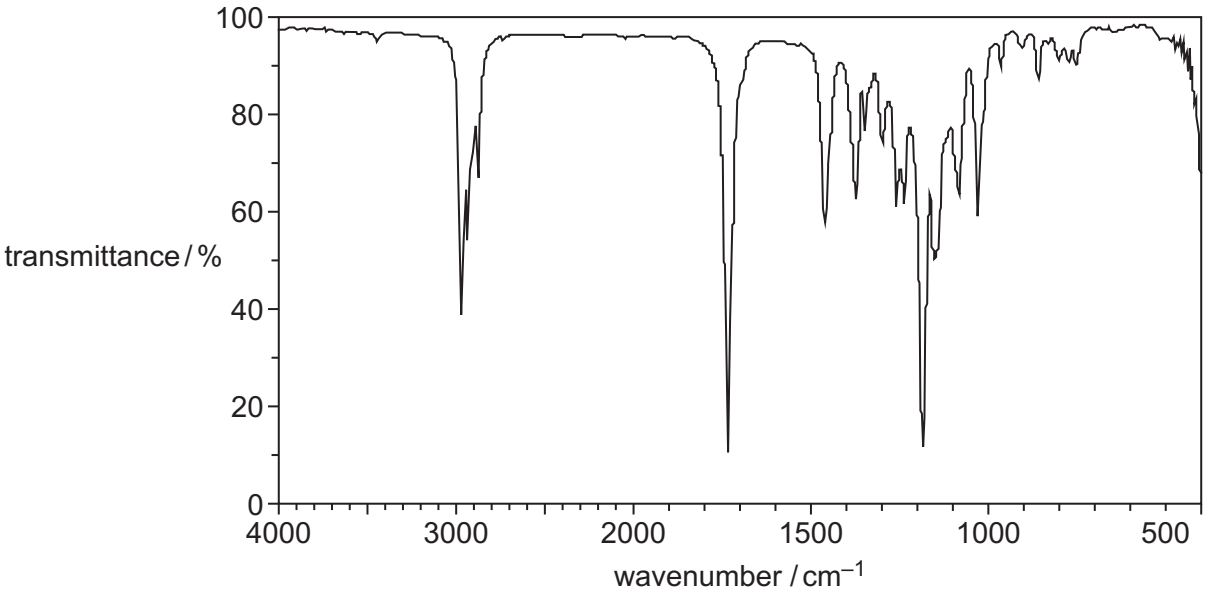


Fig. 4.2

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C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Use Fig. 4.2 and Table 4.3 to identify the functional group present in **U**.

Explain your answer fully.

functional group .....

explanation .....

.....

.....

.....

[2]

(iii) **T** also has a single functional group.

Use the information in **(c)(i)** and your answer to **(c)(ii)** to identify **T** and **U**.

Draw the structures of **T** and **U** in the boxes.

T

U

[2]

2 Barium hydroxide, $Ba(OH)_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $Ba(OH)_2$.

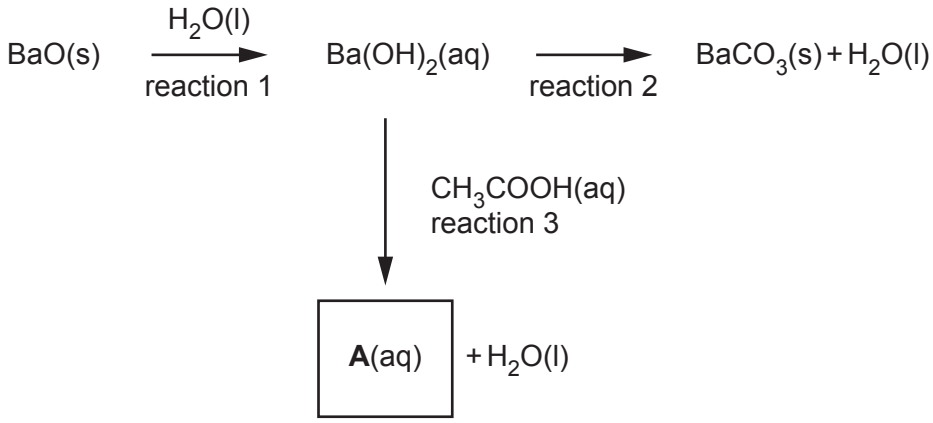


Fig. 2.1

(a) (i) State the variation in solubilities of group 2 hydroxides.

..... [1]

(iv) Identify **A**.

..... [1]

(c) $Ba(OH)_2$ is used to hydrolyse organic compounds.

Fig. 2.2 shows the reaction of **B** with $Ba(OH)_2$, followed by acidification.

Draw the structures of the organic products of the process shown in Fig. 2.2.

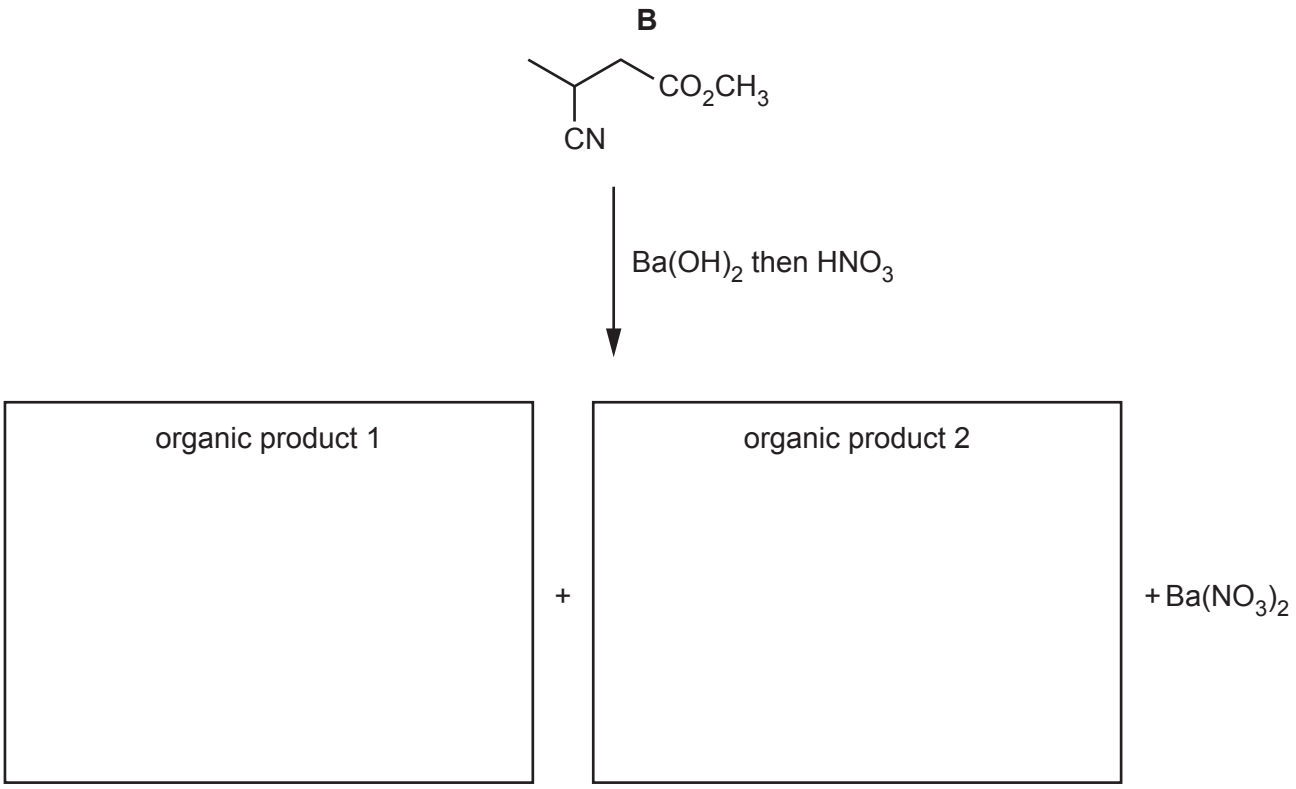


Fig. 2.2

[3]

3 Phosphoric(V) acid, H_3PO_4, is used in both inorganic and organic reactions.

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4.

(i) Write an equation for each step.

step 1 .....

step 2 .....

[2]

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

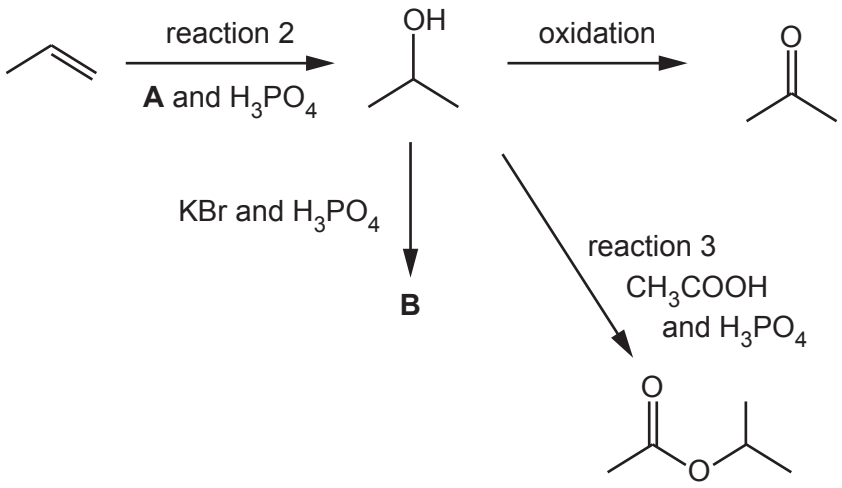


Fig. 3.1

(i) Identify **A**, which reacts with propene in the presence of H_3PO_4 in reaction 2.

..... [1]

(iii) Name the type of reaction that occurs in reaction 3.

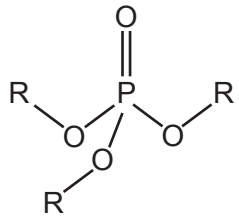
..... [1]

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C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C–H	alkane	2850–2950

(d) H_3PO_4 also reacts with alcohols to form organophosphates.

Organophosphates are compounds similar to esters. They have the general structure shown in Fig. 3.2.



R = alkyl group

Fig. 3.2

(i) Complete the equation to suggest the products of the reaction of H_3PO_4 with methanol, CH_3OH.

$H_3PO_4 + 3CH_3OH \rightarrow$ ..... [1]

2 Barium hydroxide, $Ba(OH)_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $Ba(OH)_2$.

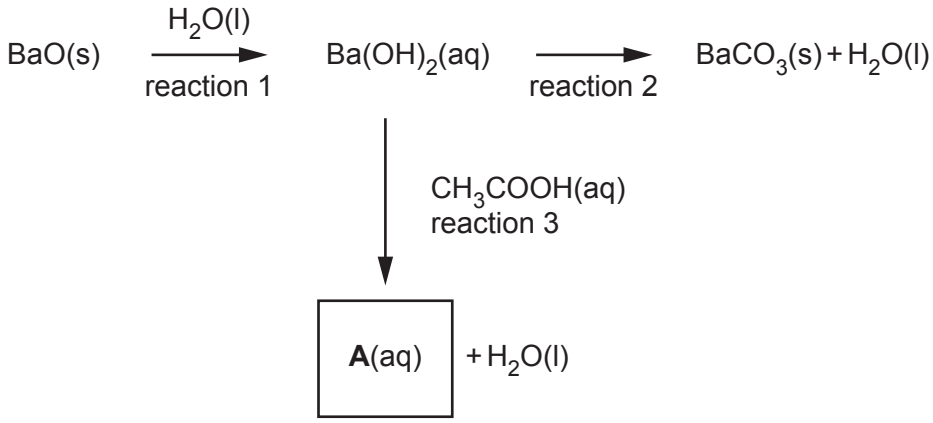


Fig. 2.1

(a) (i) State the variation in solubilities of group 2 hydroxides.

..... [1]

(iv) Identify **A**.

..... [1]

(c) $Ba(OH)_2$ is used to hydrolyse organic compounds.

Fig. 2.2 shows the reaction of **B** with $Ba(OH)_2$, followed by acidification.

Draw the structures of the organic products of the process shown in Fig. 2.2.

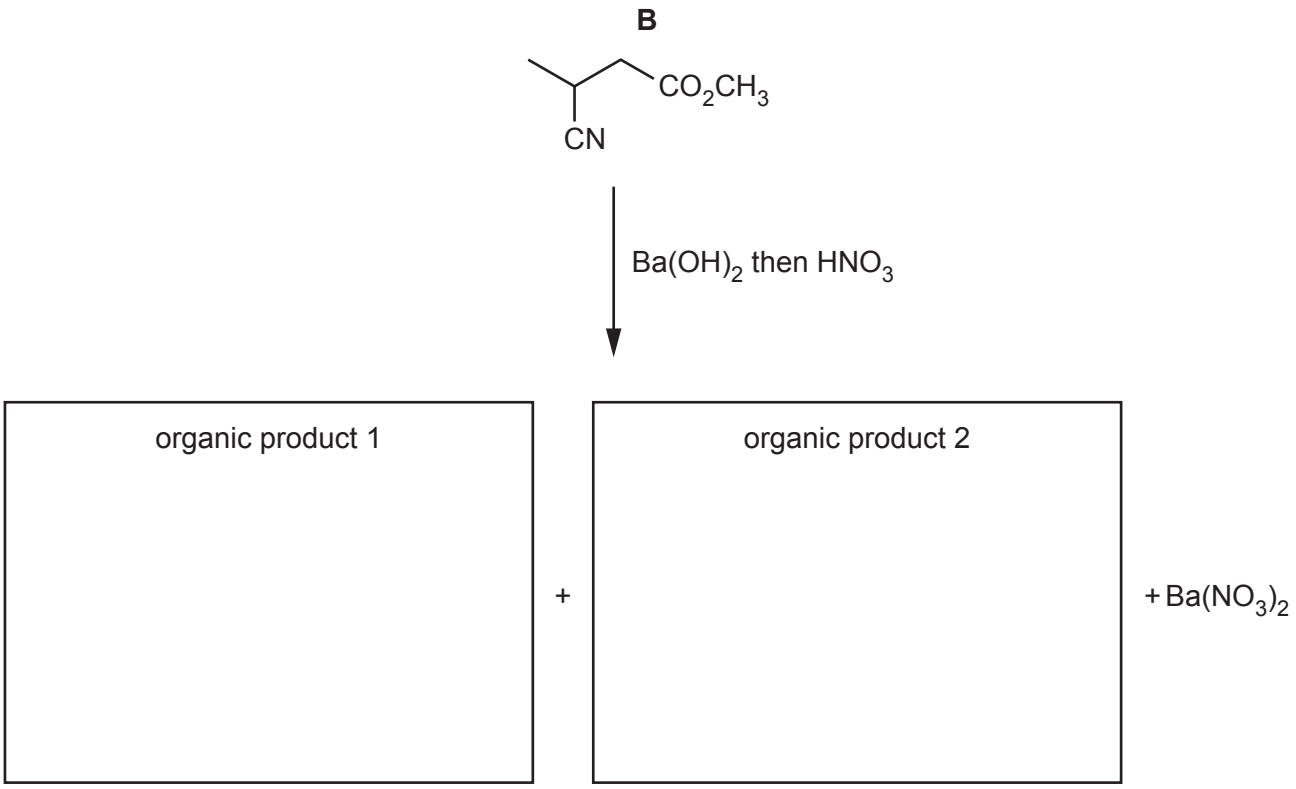


Fig. 2.2

[3]

4 Compounds **C** and **D** are alkenes with the same molecular formula, C_5H_{10}.



Fig. 4.1

(a) **(i)** Give the systematic name of **D**.

..... [1]

(c) **C** and **D** both react with HBr.

(i) **C** reacts with HBr to form **E**.

Complete the diagram in Fig. 4.2 to show the mechanism for this reaction.

Draw the structure of the organic intermediate.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.2

[3]

(d) **C** can be used to form **H**.

H

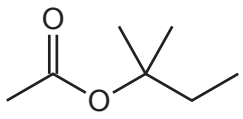


Fig. 4.4

One possible synthesis of **H** is shown in Fig. 4.5. Different portions of **C** are used in reactions 1 and 3. Some of the products are then combined to produce **H**.

Fig. 4.5 does not show any of the inorganic products of the reactions.

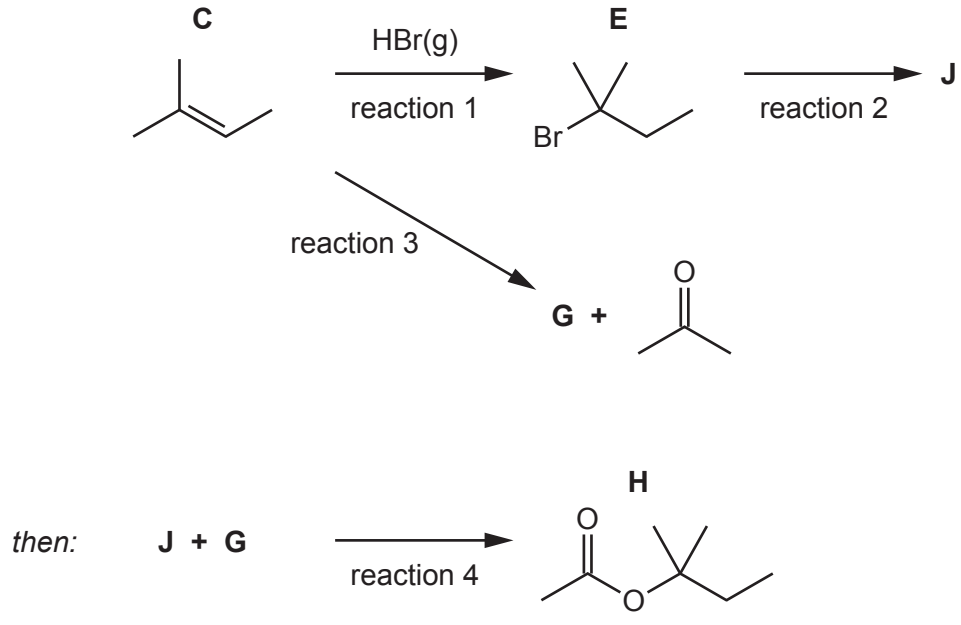


Fig. 4.5

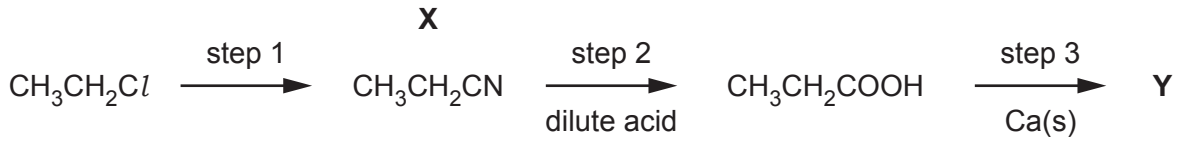
Complete Table 4.1 with the reagents and conditions required for each of the reactions shown in Fig. 4.5.

Table 4.1

	reagent and conditions
reaction 1 **C** → **E**	HBr(g)
reaction 2 **E** → **J**	
reaction 3 **C** → **G** + 	
reaction 4 **J** + **G** → **H**	

[3]

3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from CH_3CH_2Cl.



CH_3CH_2Cl → CH_3CH_2CN → CH_3CH_2COOH

Fig. 3.1

(a) (i) State the reagent and conditions for step 1 in Fig. 3.1.

..... [1]

(iv) In step 3, **Y** and a gas are produced.
Construct an equation for step 3.

..... [2]

(b) CH_3CH_2COOH can also be formed from propan-1-ol and potassium dichromate(VI).
State the conditions required.

..... [1]

5 **(a)** Describe structural isomerism.

.....

..... [1]

(d) **D** reacts in the presence of a sulfuric acid catalyst to form **E** and water.

The structure of **E** is shown in Fig. 5.1.

E

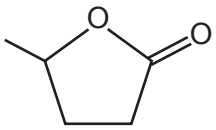


Fig. 5.1

(i) Name the functional group present in **E**.

..... [1]

(ii) Identify the type of reaction that occurs when **D** reacts to form **E**.

..... [1]

(iii) Draw the structure of **D** in the box.

D

[1]

4 **V** is a colourless liquid.

V

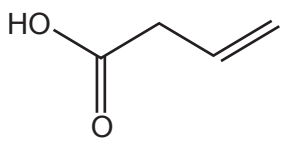


Fig. 4.1

(a) **V** reacts with an excess of $LiAlH_4$ to form **W**.

(i) Draw the structure of **W** in the box.

W

[1]

(ii) Identify the role of $LiAlH_4$ in the reaction with **V**.

..... [1]

5 **(a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.

(i) Construct an equation for the complete combustion of **M**.

..... [2]

(c) Biodiesel **T** is a fuel made from vegetable oil **R**. Fig. 5.1 shows the production of **T** from **R** in a two-step process.

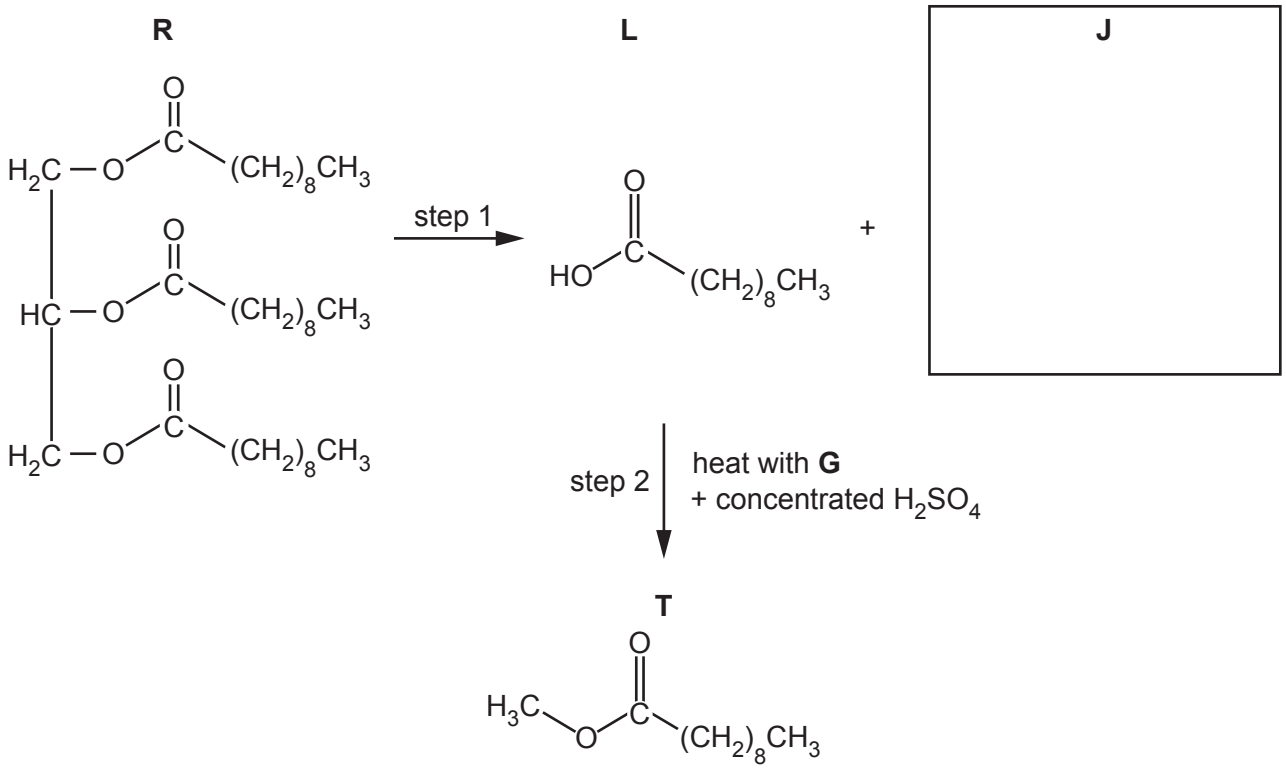


Fig. 5.1

(i) In step 1 all three ester groups in **R** react. Suggest a suitable reagent and conditions for step 1.

..... [1]

(ii) Draw the structural formula of **J** in the box in Fig. 5.1. [1]

(iii) Name the type of reaction that occurs in step 2.

..... [1]

(iv) Name organic reagent **G** used in step 2.

..... [1]

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

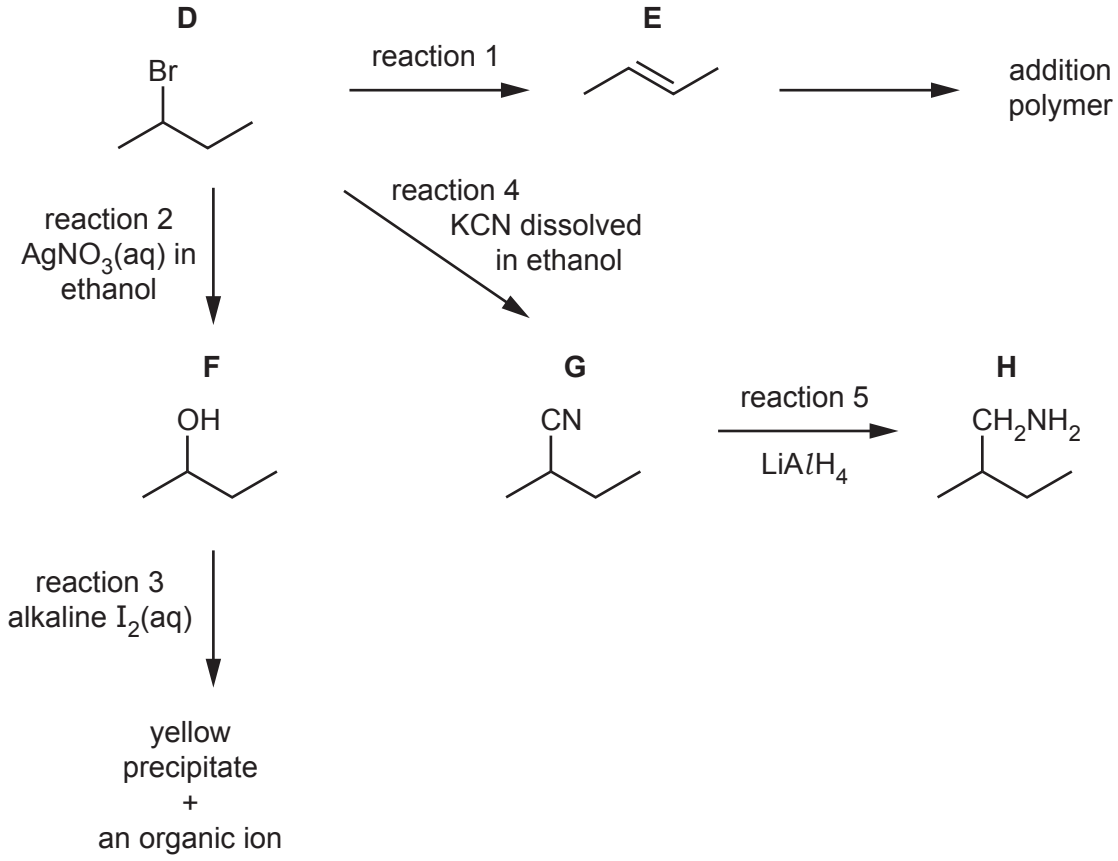


Fig. 4.1

(a) (i) State the reagent and conditions used to form **E** in reaction 1.

..... [1]

(c) (i) State the type of reaction that occurs in reaction 4.

..... [1]

(ii) Reaction 5 is similar to the reaction of $LiAlH_4$ with carboxylic acids to form alcohols.

Suggest the role of $LiAlH_4$ in reaction 5.

..... [1]